

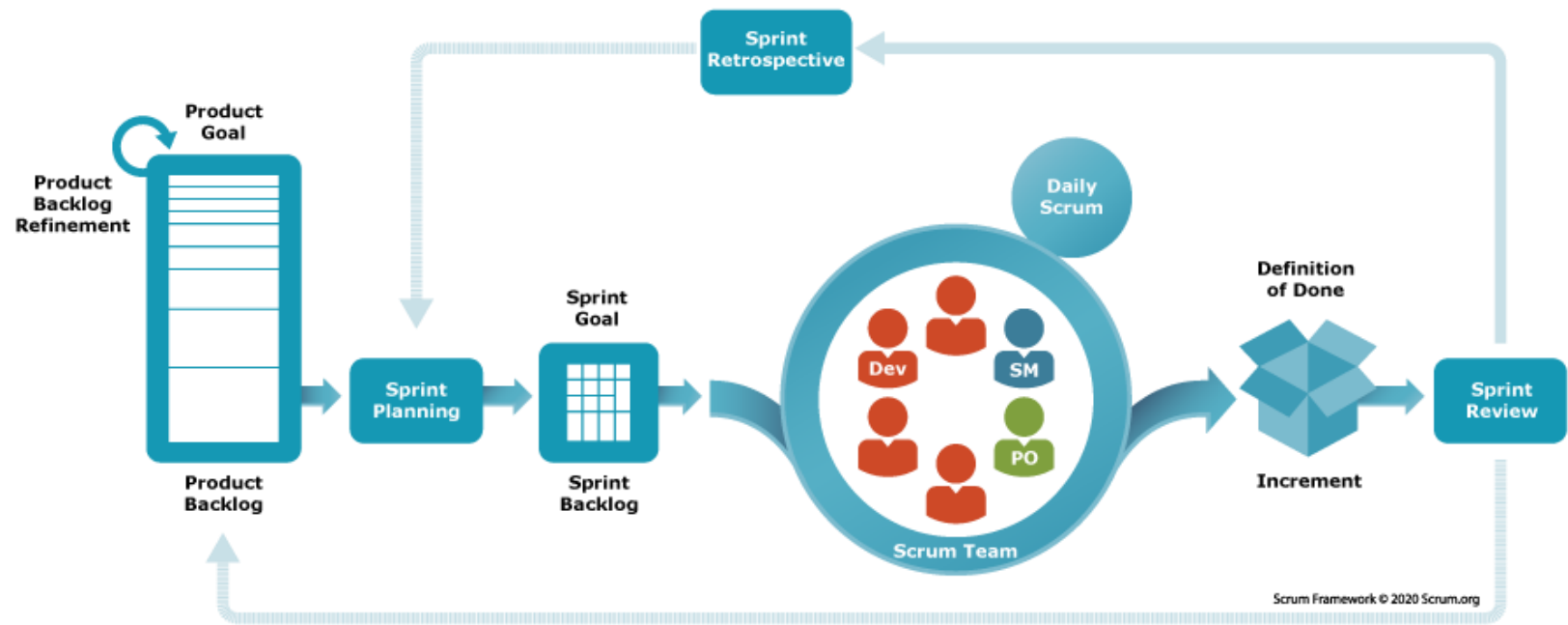
Back End in Java for Information Processing

DEVELOP - SPRINT 1

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What is SCRUM?

Scrum helps people and teams deliver value incrementally in a collaborative way. As an agile framework, Scrum provides just enough structure for people and teams to integrate into how they work, while adding the right practices to optimize for their specific needs.



Roles

- **Product Owner** → Defines the vision, manages the Product Backlog, and prioritizes work.
- **Scrum Master** → Facilitates Scrum, removes impediments, and ensures the team follows Scrum practices.
- **Development Team** → Cross-functional professionals who build the product.

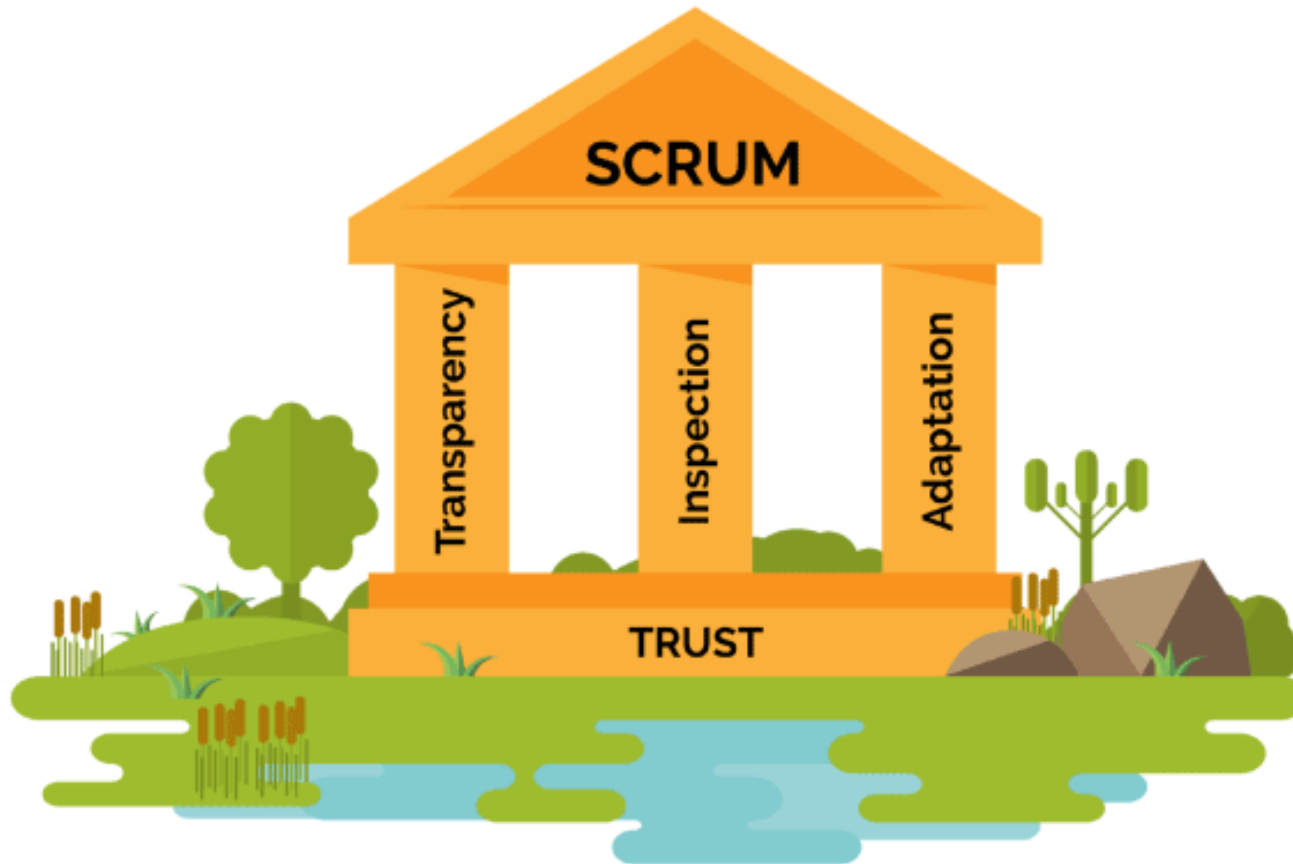
Artifacts

- **Product Backlog** → Ordered list of everything needed in the product.
- **Sprint Backlog** → Items selected for the Sprint + plan to deliver them.
- **Increment** → The working product delivered at the end of each Sprint.

Events (Ceremonies)

- **Sprint** → Time-boxed iteration (usually 2–4 weeks).
- **Sprint Planning** → Define Sprint Goal and select backlog items.
- **Daily Scrum** → 15-minute daily meeting to synchronize and plan next 24h.
- **Sprint Review** → Inspect increment and adapt Product Backlog.
- **Sprint Retrospective** → Reflect and improve the team's process.

Scrum Values



COURAGE

Scrum Team members have courage to do the right thing and work on tough problems



FOCUS

Everyone focuses on the work of the Sprint and the goals of the Scrum Team



COMMITMENT

People personally commit to achieving the goals of the Scrum Team



RESPECT

Scrum Team members respect each other to be capable, independent people



OPENNESS

The Scrum Team and its stakeholders agree to be open about all the work and the challenges with performing the work

Credit: ABN AMRO Bank N.V.

Scrum Characteristics:

- **Self-Organized** – Los equipos se organizan solos y toman decisiones.
- **Empowered** – Tienen autonomía y recursos para cumplir objetivos.
- **Collaboration** – Se fomenta trabajo en conjunto, aprendizaje y experimentación.
- **Shared Goal** – Todo el equipo trabaja hacia un objetivo común.
- **Optimum Size** – Equipos pequeños (6–10) para comunicación efectiva.
- **Diverse Skills** – Equipos multifuncionales con distintas habilidades.
- **Collocated** – Idealmente trabajan juntos físicamente, aunque se adapta a remoto.

JSON AND CSV

JSON

JSON is a data exchange format that stands for JavaScript Object Notation. JSON is known as a light-weight format type and is favored for its readability and nesting features. It is often used in conjunction with APIs and data configuration. More specifically, JSON is preferred for API use, which prioritizes file size due to its lightweight feature. JSON is also praised for its ability to be easily converted into other formats. JSON conversion tools are quite common, and there are many free JSON to CSV format conversion tools out there.

CSV

CSV is a data storage format that stands for Comma Separated Values. Implied in its name, CSV stores the data (values) in a list format separated by commas. CSV is noted for its small file size and simplicity. Likewise, because of its simplicity, CSV can be used by virtually anyone who is tasked with examining simple data in spreadsheets and tables. CSV files can be converted to JSON format, but complex JSON files may lead to reading and writing errors.

JSON

```
{  
  "name": "John Doe",  
  "age": 30,  
  "skills": ["JavaScript", "Python", "SQL"],  
  "address": {  
    "city": "New York",  
    "zip": "10001"  
  }  
}
```

CSV

```
"Name", "Age", "Skills", "City"  
"John Doe", "30", "JavaScript;Python;SQL", "New York"  
"Jane Smith", "25", "HTML;CSS", "Los Angeles"
```

JSON

(JavaScript Object Notation)

vs

CSV

(Comma Separated Values)

File size

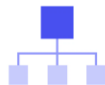
Larger file size



Compact file size

Hierarchy

Supports hierarchical and relational data



Errors when displaying hierarchical data

Scalability

Allows scalability and integrates with APIs easily



Not easily scalable and difficult to integrate

Best for

Works best for complex and large-scale datasets



Convenient for small datasets

XML

vs.

JSON

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <endereco>
3   <cep>31270901</cep>
4   <city>Belo Horizonte</city>
5   <neighborhood>Pampulha</neighborhood>
6   <service>correios</service>
7   <state>MG</state>
8   <street>Av. Presidente Antônio Carlos, 6627</street>
9 </endereco>
```

```
1 {
2   "endereco": {
3     "cep": "31270901",
4     "city": "Belo Horizonte",
5     "neighborhood": "Pampulha",
6     "service": "correios",
7     "state": "MG",
8     "street": "Av. Presidente Antônio Carlos, 6627"
9   }
10 }
```